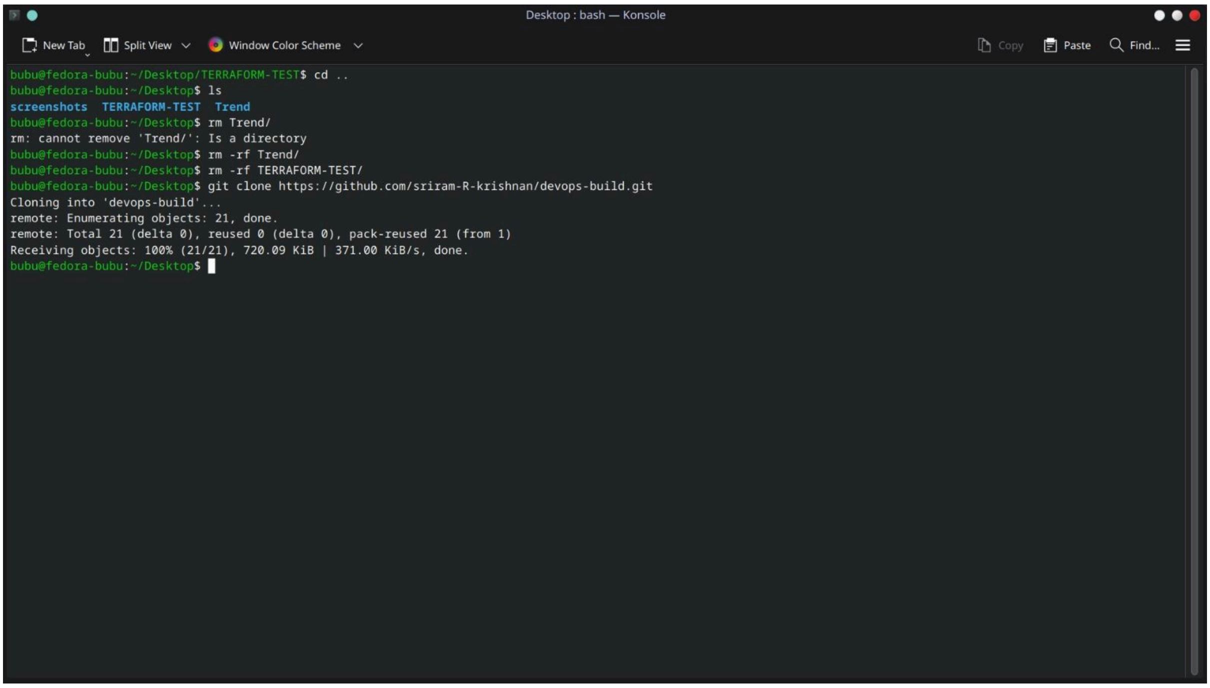
# Reactjs E-commerce Application Documentation

## Step 1 : Cloning the repository

- Git clone https://github.com/sriram-R-krishnan/devops-build

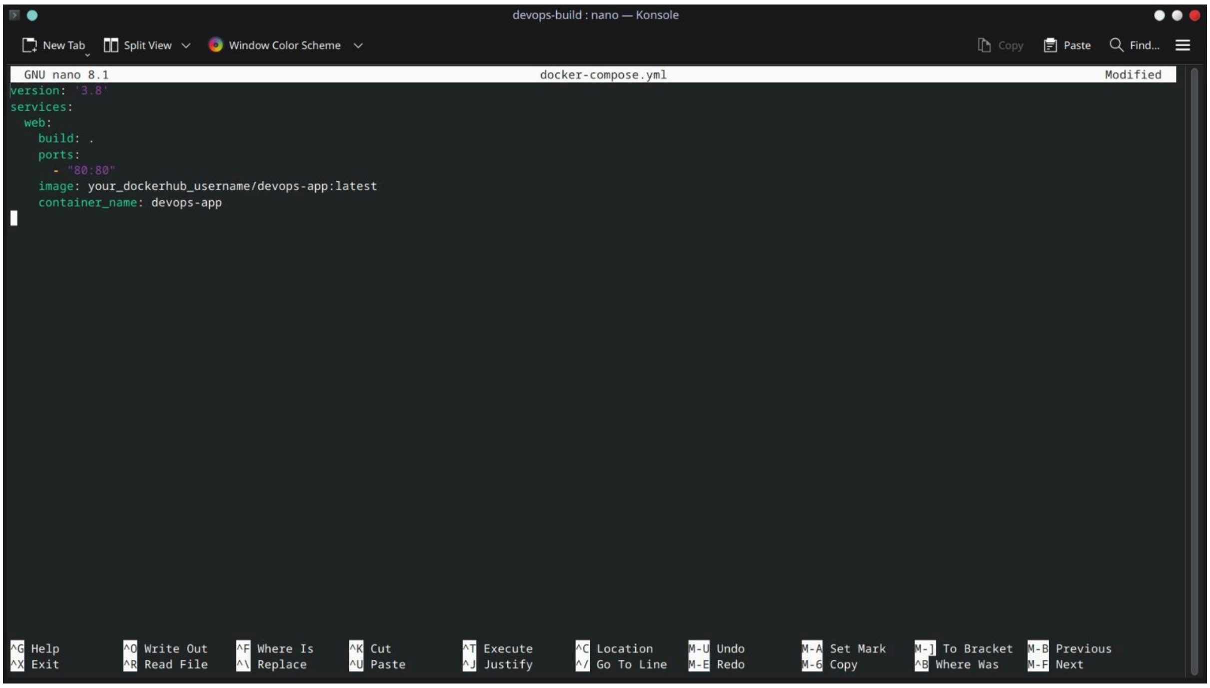


## Step 2 : Dockerize the application using dockerfile

FROM nginx:stable-alpine
COPY ./build /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]

## Step 3 : docker-compose file

```yaml
version: '3.8'
services:
  web:
    build: .
    ports:
      - "80:80"
    image: your_dockerhub_username/devops-app:latest
    container_name: devops-app
```

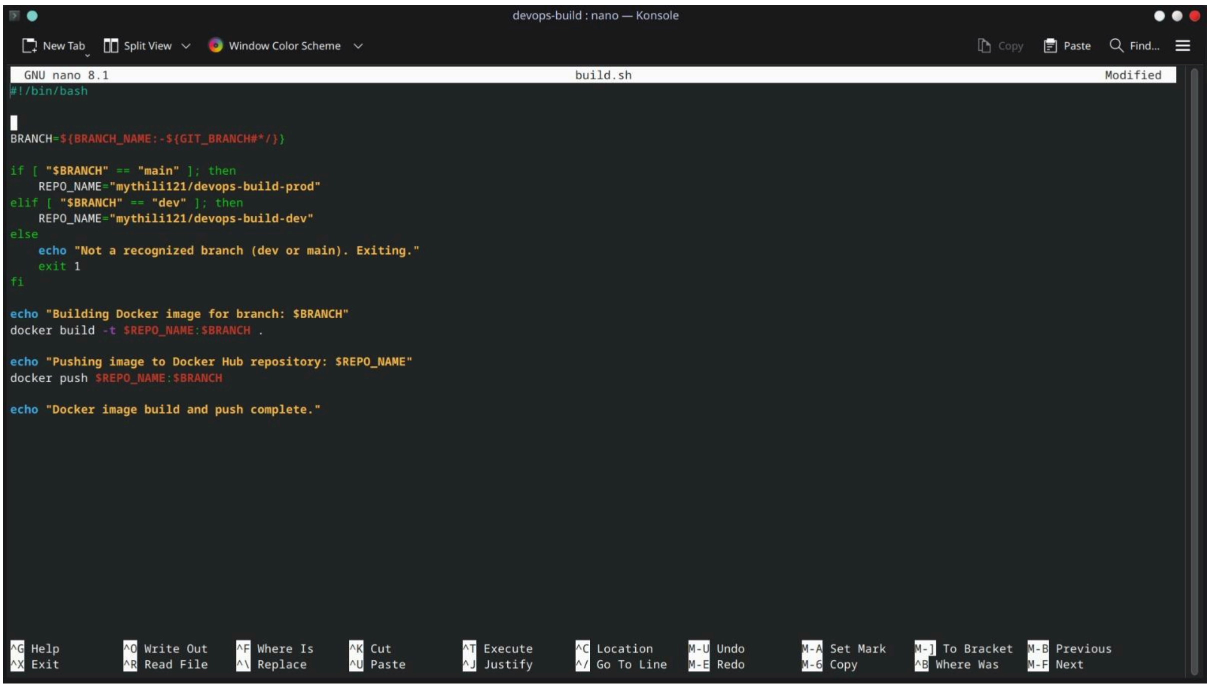


## Step 4 : Bash scripting

- Create build.sh and deploy.sh and make it executable

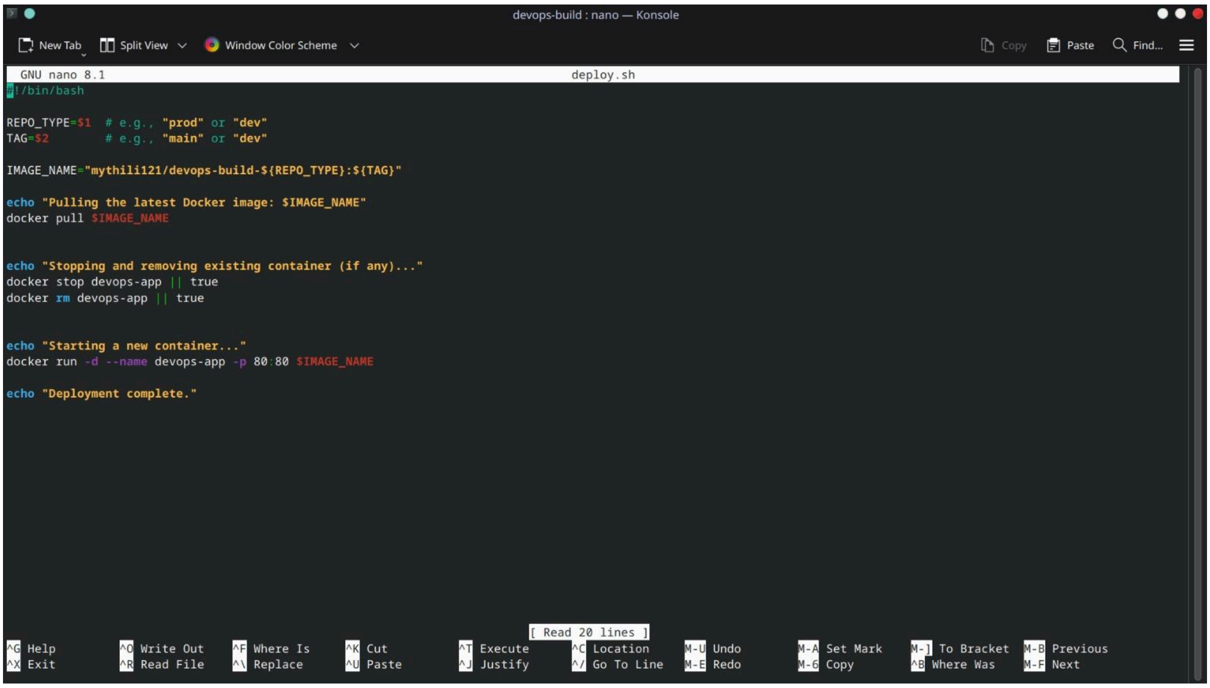
build.sh

```
#!/bin/bash

# Prefer BRANCH_NAME if set (Multibranch), otherwise fallback to GIT_BRANCH
BRANCH=${BRANCH_NAME:-${GIT_BRANCH#*/}}

if [ "$BRANCH" == "main" ]; then
  REPO_NAME="mythili121/devops-build-prod"
elif [ "$BRANCH" == "dev" ]; then
 REPO_NAME="mythili121/devops-build-dev"
else
 echo "Not a recognized branch (dev or main). Exiting."
  exit 1
fi

echo "Building Docker image for branch: $BRANCH"
docker build -t $REPO_NAME:$BRANCH .

echo "Pushing image to Docker Hub repository: $REPO_NAME"
docker push $REPO_NAME:$BRANCH

echo "Docker image build and push complete."
```



deploy.sh

```
#!/bin/bash

REPO_TYPE=$1  # e.g., "prod" or "dev"
TAG=$2        # e.g., "main" or "dev"

IMAGE_NAME="mythili121/devops-build-${REPO_TYPE}:${TAG}"

echo "Pulling the latest Docker image: $IMAGE_NAME"
docker pull $IMAGE_NAME


echo "Stopping and removing existing container (if any)..."
docker stop devops-app || true
docker rm devops-app || true


echo "Starting a new container..."
docker run -d --name devops-app -p 80:80 $IMAGE_NAME

echo "Deployment complete."
```

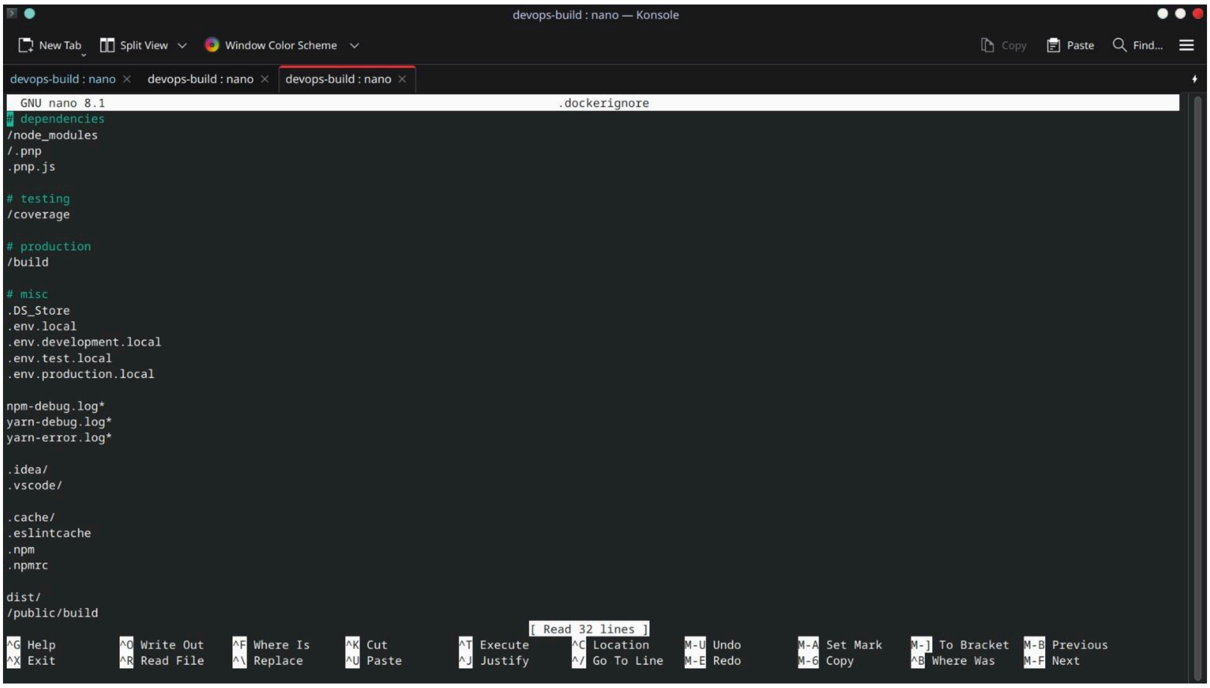


## Step 5 : Version Control

- Work on features locally, commit and push to the dev branch.
- dev branch: continuous integration builds and pushes to Docker Hub dev repo.

<u>Dockerignore:</u>
.git
.gitignore
.dockerignore
Node_modules
Npm-debug.log

<u>README.md</u>
**/.*
.vscode
.idea




<u>gitignore:</u>
# dependencies
/node_modules
/.pnp
.pnp.js

# testing
/coverage

# production

/build

# misc
.DS_Store
.env.local
.env.development.local
.env.test.local
.env.production.local

npm-debug.log*
yarn-debug.log*
yarn-error.log*

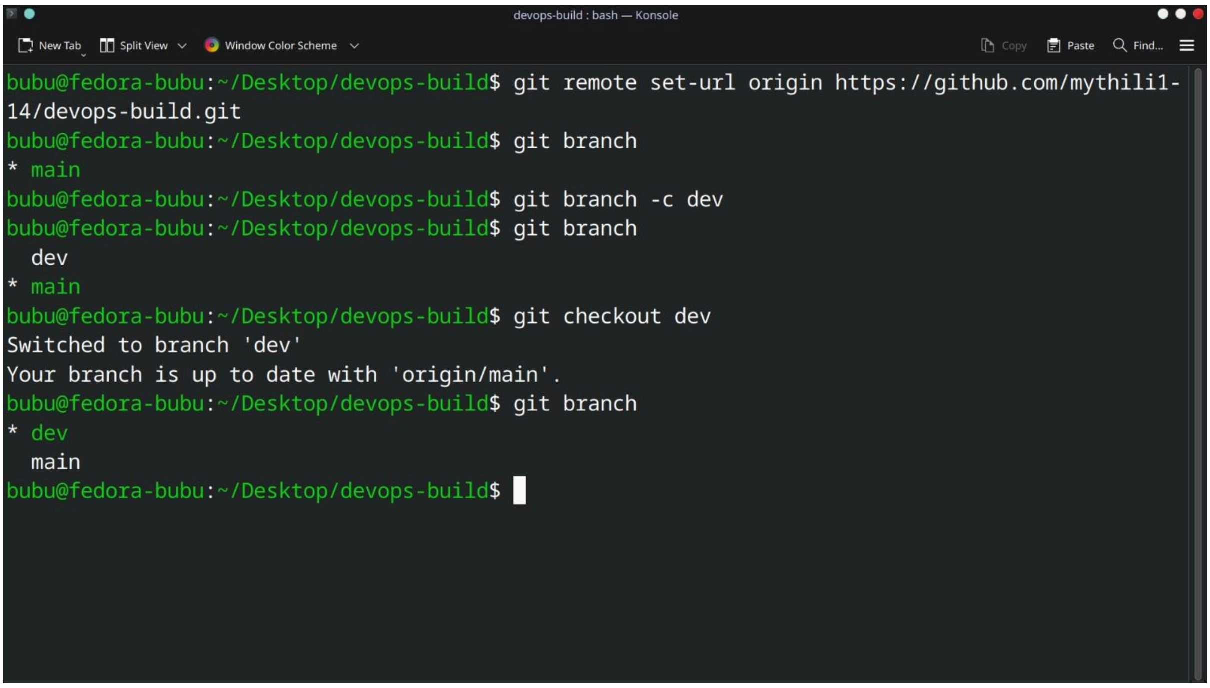
.idea/
.vscode/

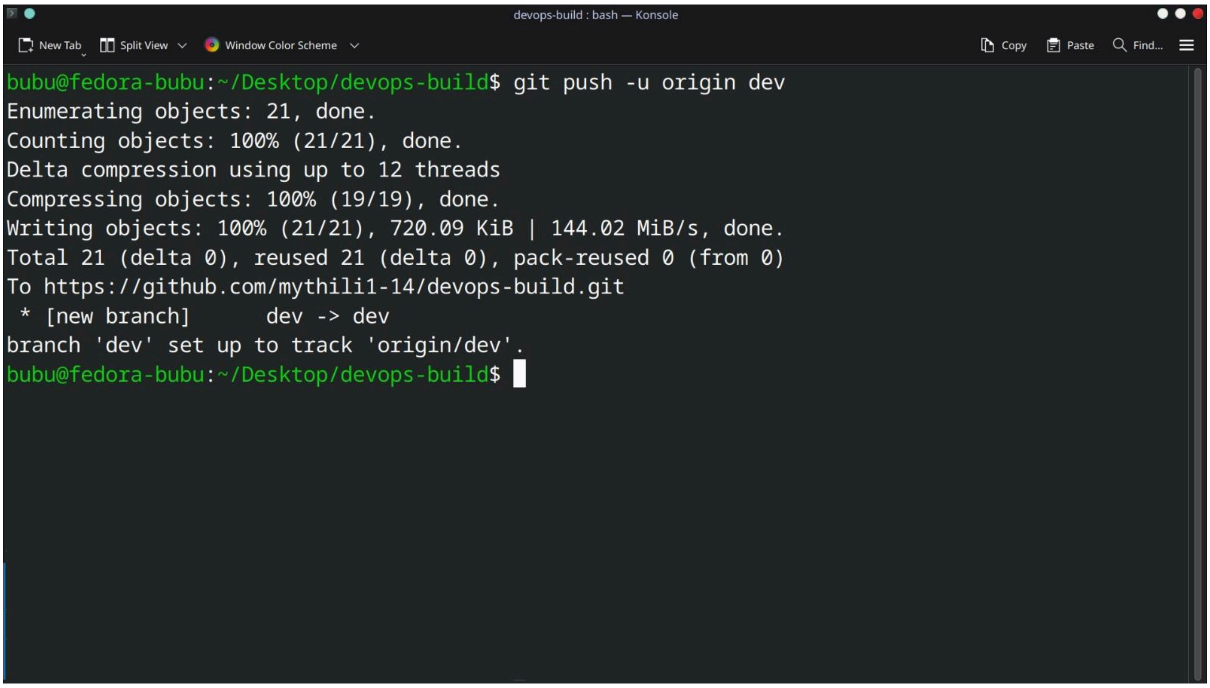
.cache/
.eslintcache
.npm
.npmrc

dist/
/public/build

Git commands

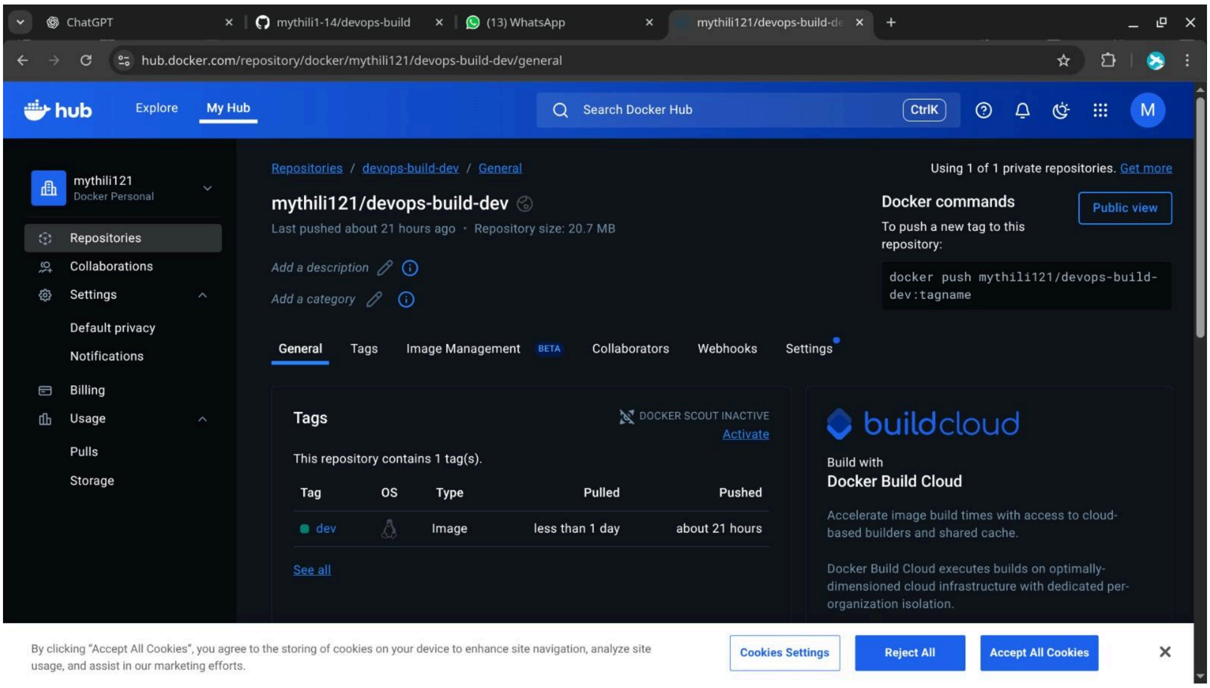
- git add .
- git commit -m "feat: add X"
- git push origin dev
- # When merged to master:
- git checkout master
- git pull origin master

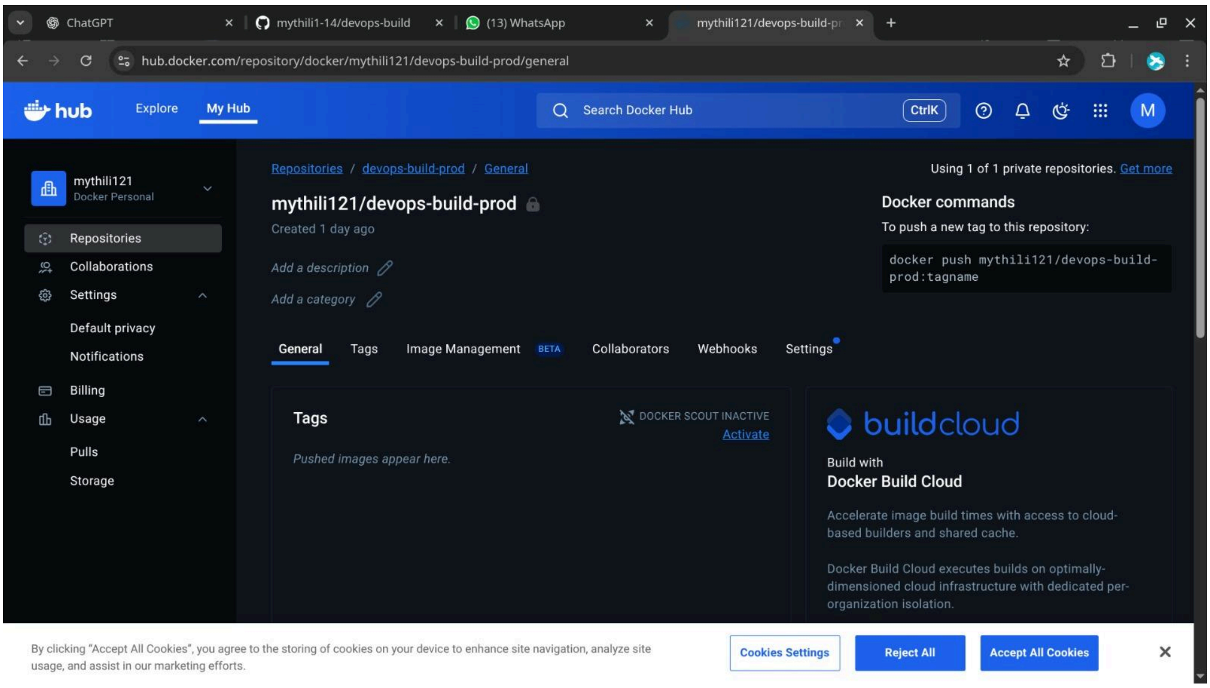


## Step 6 : Docker hub:

- Create two repositories:
  - devops-build-dev — public
  - devops-build-prod — private
- Use docker tag and docker push to upload images

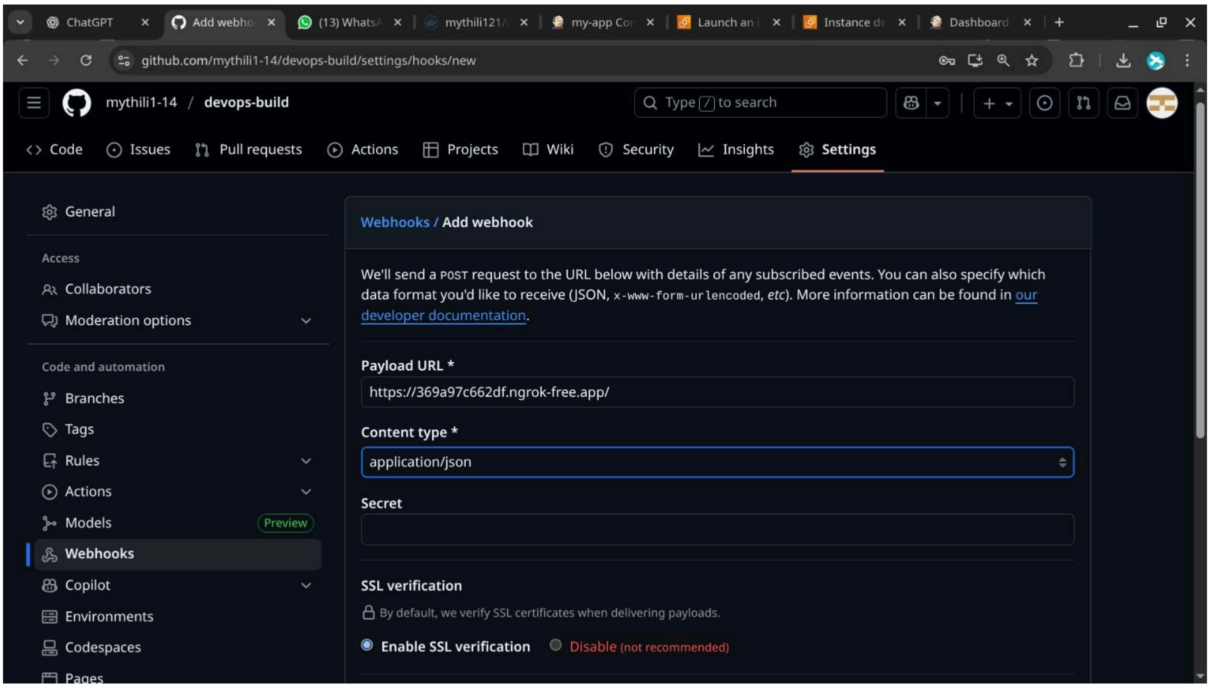


```
nothing added to commit but untracked files present (use "git add" to track)
bubu@fedora-bubu:~/Desktop/devops-build$ git add .
bubu@fedora-bubu:~/Desktop/devops-build$ git commit -m "initial commit" .
[dev c061fc1] initial commit
 6 files changed, 120 insertions(+)
 create mode 100644 .dockerignore
 create mode 100644 .gitignore.save
 create mode 100644 Dockerfile
 create mode 100644 build.sh.save
 create mode 100644 deploy.sh
 create mode 100644 docker-compose.yml
bubu@fedora-bubu:~/Desktop/devops-build$ git push origin dev
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 12 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (7/7), 1.37 KiB | 1.37 MiB/s, done.
Total 7 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mythili1-14/devops-build.git
   ef44730..c061fc1  dev -> dev
bubu@fedora-bubu:~/Desktop/devops-build$
```

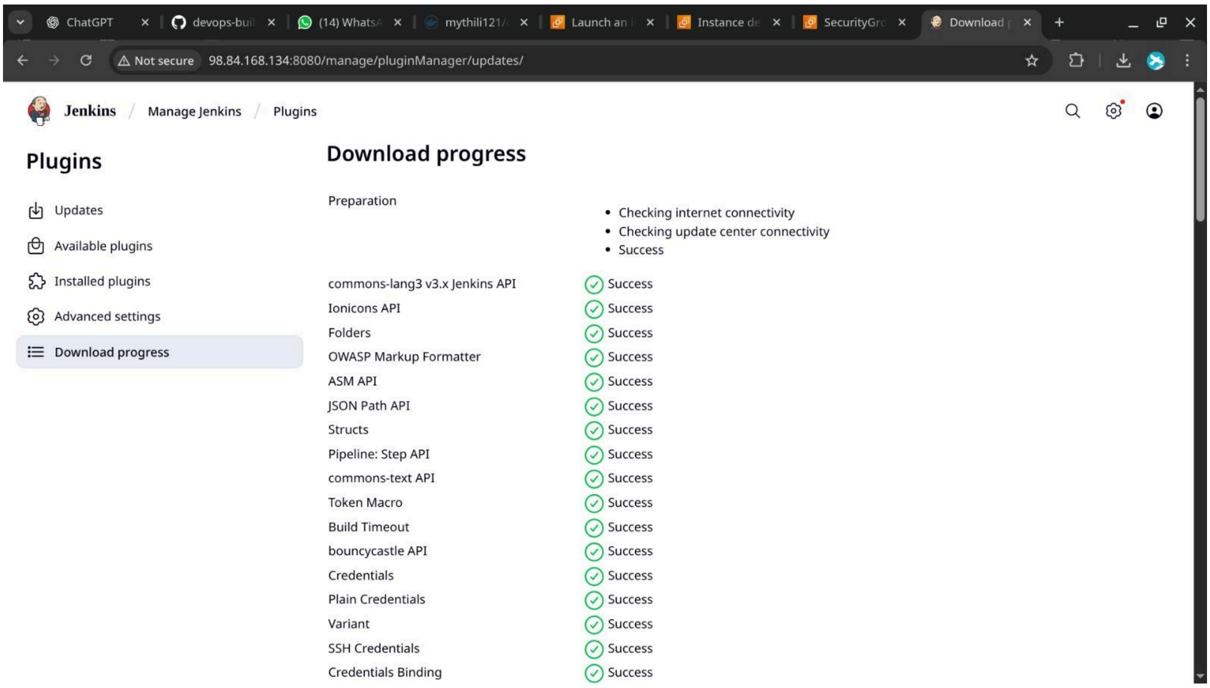
## Step 7 : Jenkins

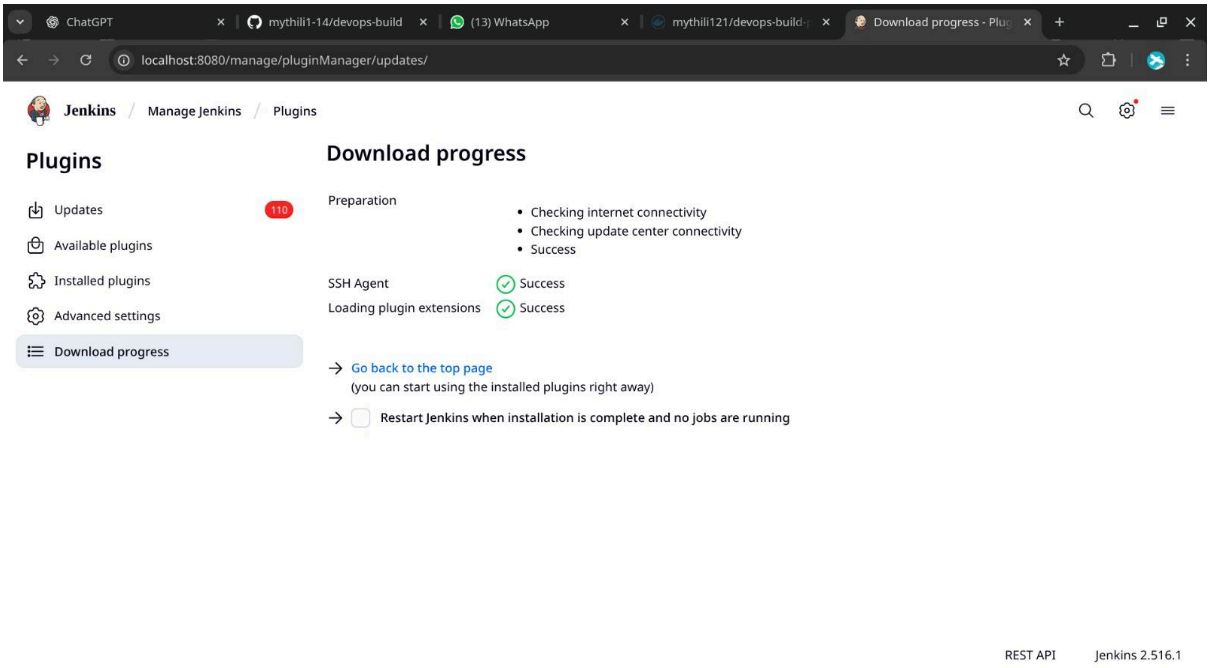
- Install Jenkins (on a server or VPS)
- Install Java 11+, install Jenkins LTS.
- Install plugins: GitHub, Docker, Pipeline, Credentials, GitHub Branch Source.
- Create credentials
    - Docker Hub credentials (username/password)
    - GitHub credentials/token (if private repo)
    - SSH key credential for EC2 (if using ssh deploy)
- Configure webhooks
    - In GitHub repo settings → Webhooks: add Jenkins webhook URL so pushing to dev or master triggers build automatically.

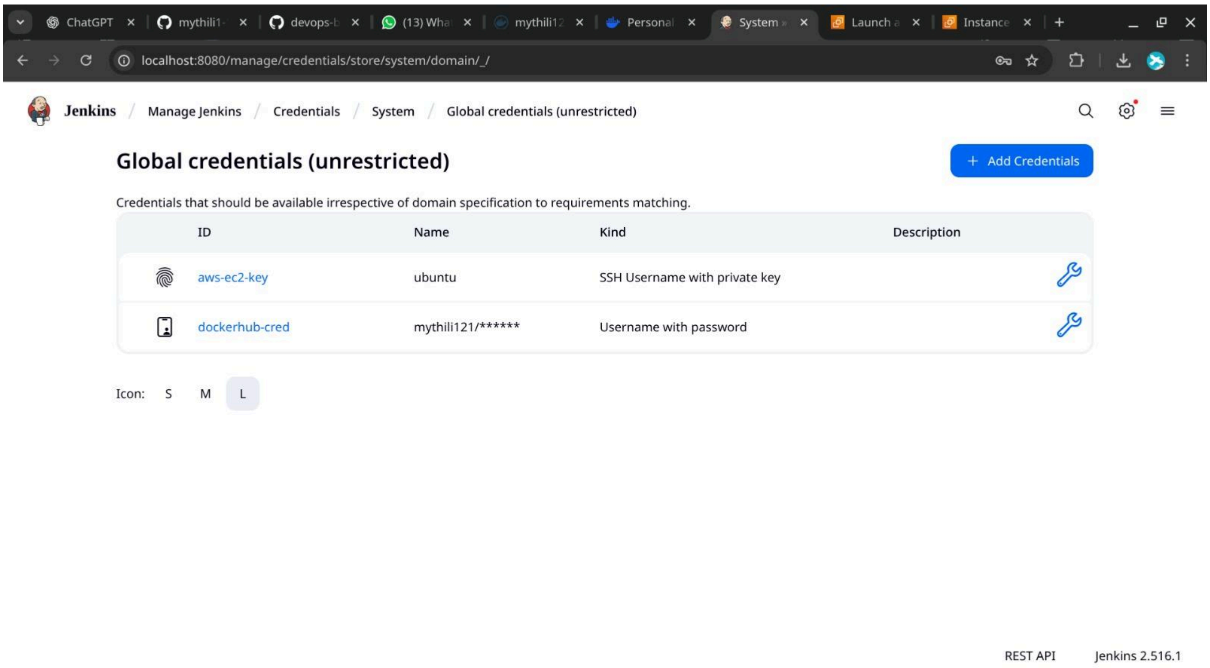


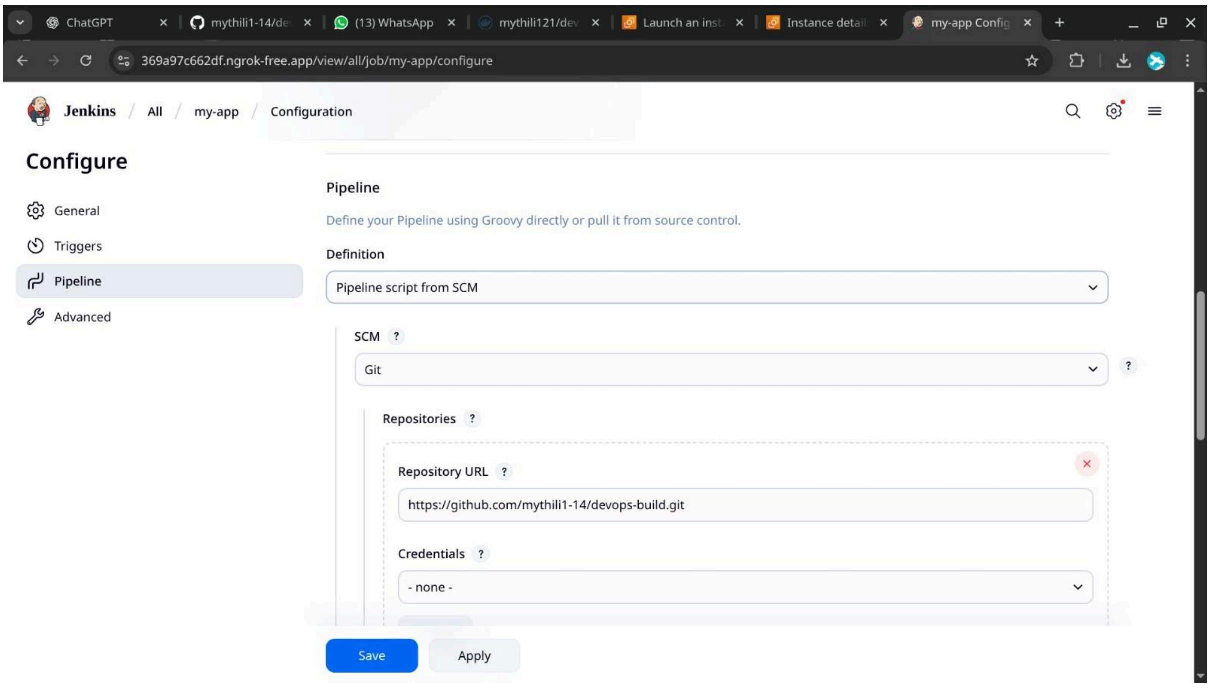
Jenkinsfile:

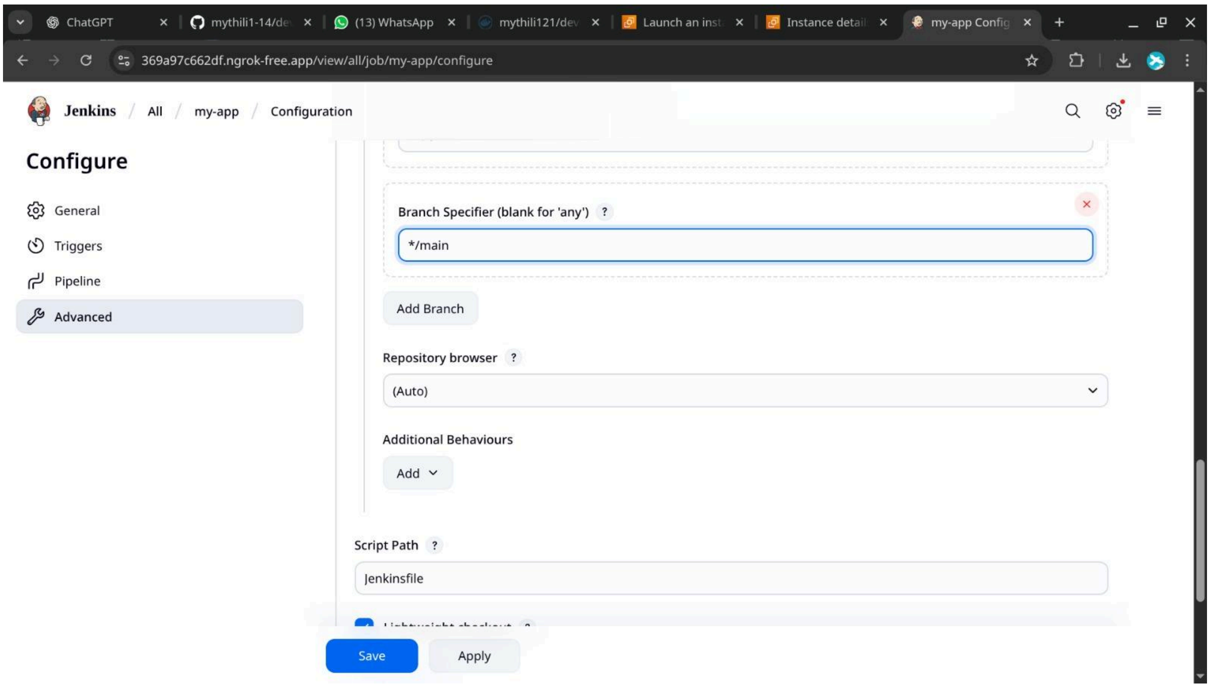
```
pipeline {
   agent any
    environment {
          DOCKERHUB_USERNAME = 'mythili121'
           DOCKERHUB_CREDENTIAL_ID = 'dockerhub-cred'
           AWS_SSH_CREDENTIAL_ID = 'aws-ec2-key'
            EC2_IP = '54.172.118.89'
      }

stages {
            stage('Build and Push') {
                     steps {
                           withCredentials([usernamePassword(credentialsId:
DOCKERHUB_CREDENTIAL_ID, usernameVariable: 'DOCKER_USERNAME',
passwordVariable: 'DOCKER_PASSWORD')]) {
          sh "echo \$DOCKER_PASSWORD | docker login -u \$DOCKER_USERNAME
--password-stdin"
               }
               sh "bash ./build.sh"
    }
  }
    stage('Deploy to Server') {
          steps {
              sshagent(credentials: [AWS_SSH_CREDENTIAL_ID]) {
              sh "scp -o StrictHostKeyChecking=no deploy.sh
ubuntu@${EC2_IP}:/tmp/deploy.sh"
               sh "ssh -o StrictHostKeyChecking=no ubuntu@${EC2_IP} bash
/tmp/deploy.sh ${env.BRANCH_NAME == 'main' ? 'prod main' : 'dev dev'}"
      }
    }
  }
 }
}
```


98.84.168.134:8080/manage/pluginManager/updates/
Jenkins / Manage Jenkins / Plugins
Plugins
Updates
Available plugins
Installed plugins
Advanced settings
Download progress
Download progress
Preparation
Checking internet connectivity
Checking update center connectivity
Success
commons-lang3 v3.x Jenkins API Success
Ionicons API Success
Folders Success
OWASP Markup Formatter Success
ASM API Success
JSON Path API Success
Structs Success
Pipeline: Step API Success
commons-text API Success
Token Macro Success
Build Timeout Success
bouncycastle API Success
Credentials Success
Plain Credentials Success
Variant Success
SSH Credentials Success
Credentials Binding Success


localhost:8080/manage/pluginManager/updates/
Jenkins / Manage Jenkins / Plugins
Plugins
Updates 110
Available plugins
Installed plugins
Advanced settings
Download progress
Download progress
Preparation
Checking internet connectivity
Checking update center connectivity
Success
SSH Agent Success
Loading plugin extensions Success
Go back to the top page
(you can start using the installed plugins right away)
Restart Jenkins when installation is complete and no jobs are running
REST API
Jenkins 2.516.1


localhost:8080/manage/credentials/store/system/domain/_/
Jenkins / Manage Jenkins / Credentials / System / Global credentials (unrestricted)
Global credentials (unrestricted)
+ Add Credentials
Credentials that should be available irrespective of domain specification to requirements matching.
ID Name Kind Description
aws-ec2-key ubuntu SSH Username with private key
dockerhub-cred mythili121/****** Username with password
Icon: S M L
REST API Jenkins 2.516.1


369a97c662df.ngrok-free.app/view/all/job/my-app/configure
Jenkins / All / my-app / Configuration
Configure
General
Triggers
Pipeline
Advanced
Pipeline
Define your Pipeline using Groovy directly or pull it from source control.
Definition
Pipeline script from SCM
SCM
Git
Repositories
Repository URL
https://github.com/mythili1-14/devops-build.git
Credentials
- none -
Save
Apply

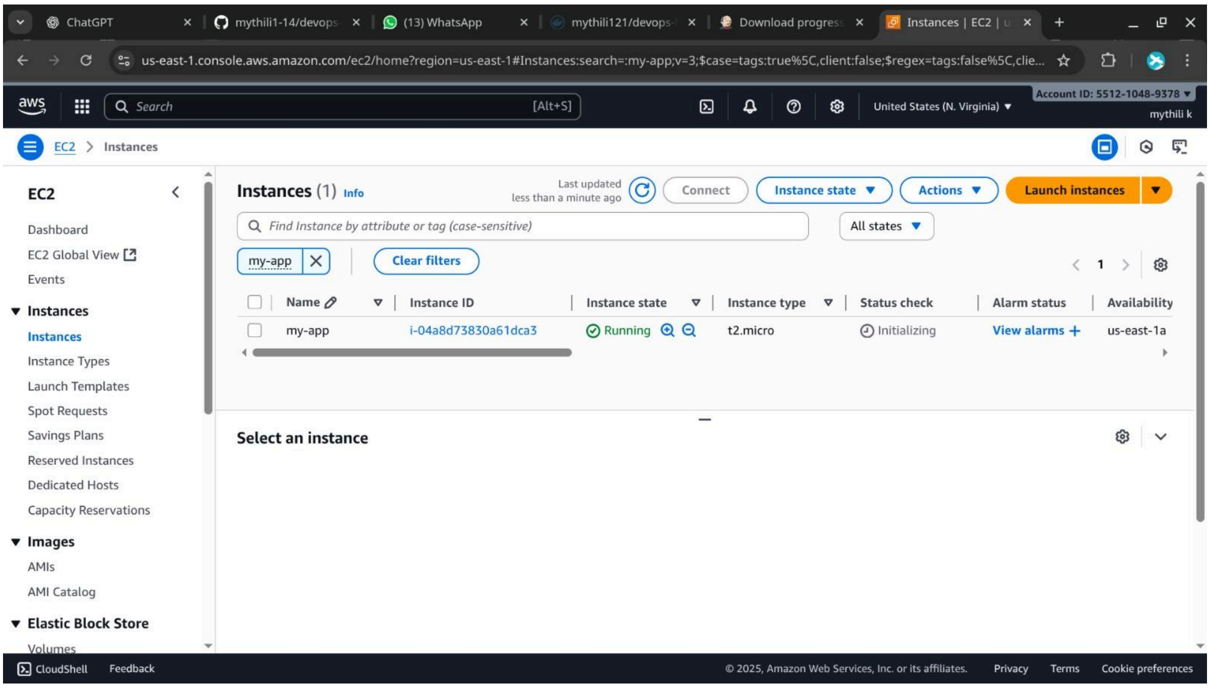

ChatGPT
mythili1-14/de
(13) WhatsApp
mythili121/dev
Launch an inst
Instance detail
my-app Config
369a97c662df.ngrok-free.app/view/all/job/my-app/configure
Jenkins
All
my-app
Configuration
Configure
General
Triggers
Pipeline
Advanced
Branch Specifier (blank for 'any')
*/main
Add Branch
Repository browser
(Auto)
Additional Behaviours
Add
Script Path
Jenkinsfile
Save
Apply

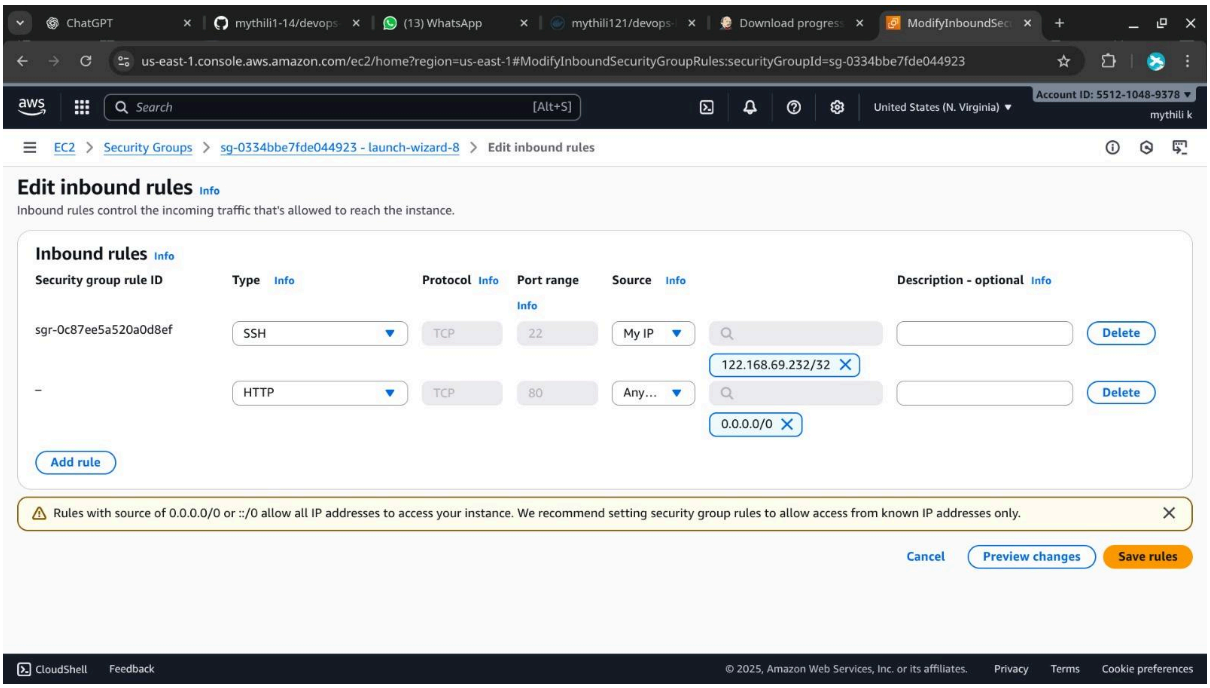
## Step 8 : AWS

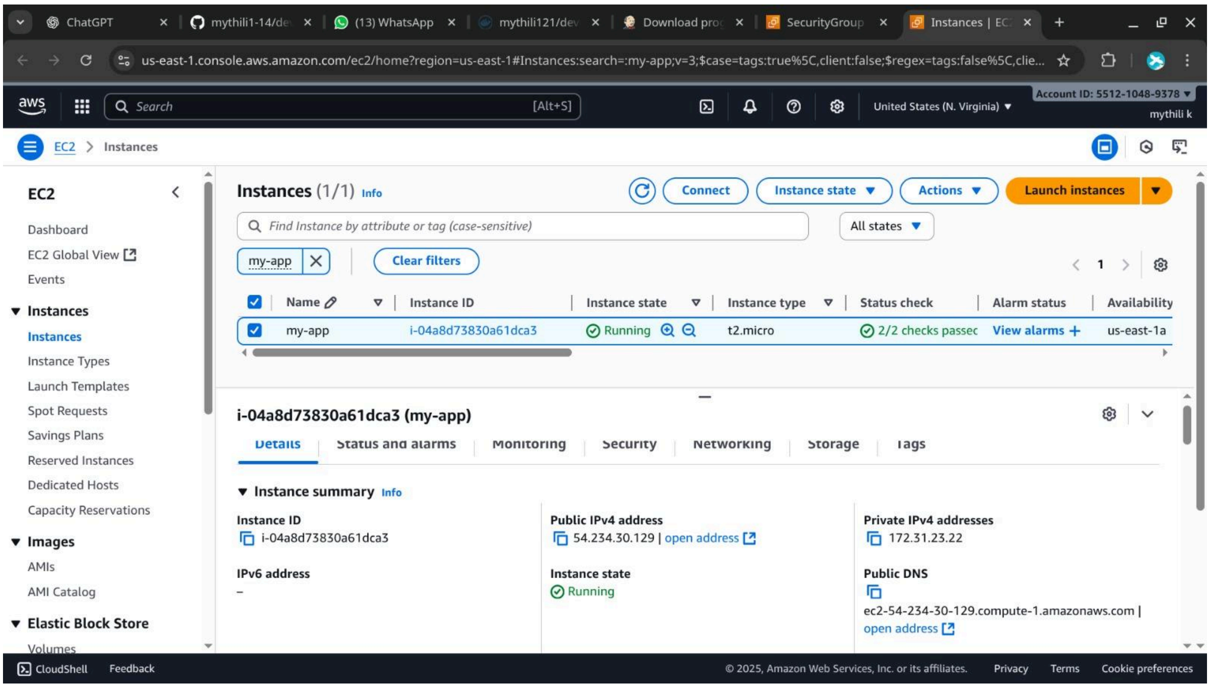
- Launch an t2.micro EC2 instance (Amazon Linux 2 or Ubuntu).
- Configure the Security Group (SG):

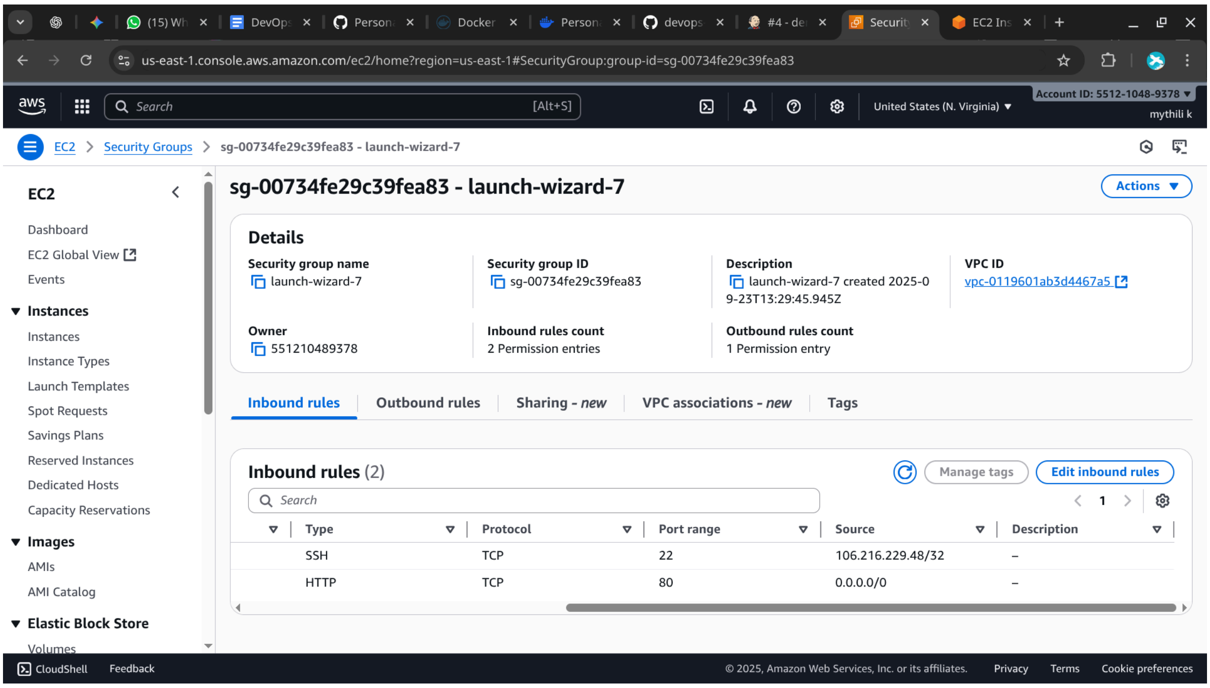
  - HTTP (TCP 80) — Source: 0.0.0.0/0 (or restrict to required IP ranges if desired).

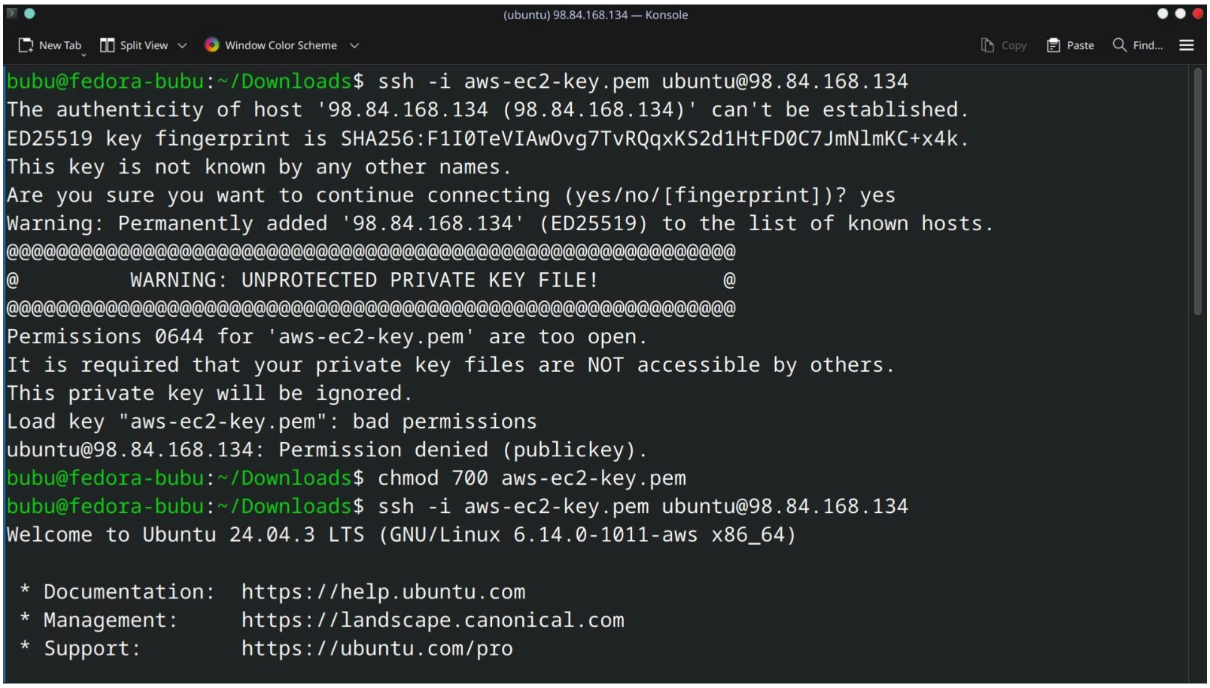
  - SSH (TCP 22) — Source: your IP only for security.
- SSH to instance and install Docker & docker-compose.
- Pull image & run using docker-compose.yml on server and docker-compose up -d.



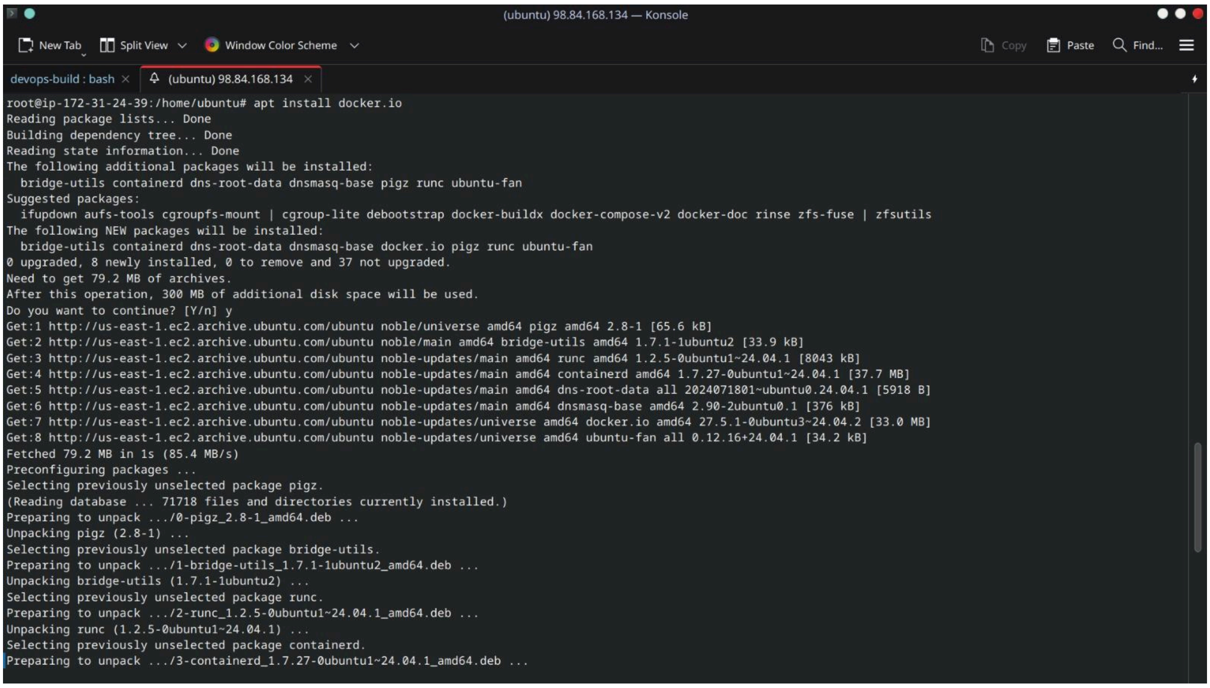




us-east-1.console.amazon.com/ec2/home?region=us-east-1#Instances:search=:my-app;v=3;$case=tags:true%5C,client:false;$regex=tags:false%5C,clie...
Account ID: 5512-1048-9378
United States (N. Virginia)
EC2 > Instances
Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
my-app
Clear filters
Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability
my-app | i-04a8d73830a61dca3 | Running | t2.micro | 2/2 checks passec | View alarms | us-east-1a
i-04a8d73830a61dca3 (my-app)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-04a8d73830a61dca3
Public IPv4 address
54.234.30.129 | open address
Private IPv4 addresses
172.31.23.22
IPv6 address
Instance state
Running
Public DNS
ec2-54-234-30-129.compute-1.amazonaws.com | open address
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


us-east-1.console.amazon.com/ec2/home?region=us-east-1#SecurityGroup:group-id=sg-00734fe29c39fea83
Account ID: 5512-1048-9378
United States (N. Virginia)
EC2 > Security Groups > sg-00734fe29c39fea83 - launch-wizard-7
sg-00734fe29c39fea83 - launch-wizard-7
Actions
Details
Security group name
launch-wizard-7
Security group ID
sg-00734fe29c39fea83
Description
launch-wizard-7 created 2025-09-23T13:29:45.945Z
VPC ID
vpc-0119601ab3d4467a5
Owner
551210489378
Inbound rules count
2 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing - new
VPC associations - new
Tags
Inbound rules (2)
Manage tags
Edit inbound rules
Type | Protocol | Port range | Source | Description
SSH | TCP | 22 | 106.216.229.48/32 | –
HTTP | TCP | 80 | 0.0.0.0/0 | –
CloudShell
Feedback
© 2025, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences




```
bubu@fedora-bubu:~/Downloads$ ssh -i aws-ec2-key.pem ubuntu@98.84.168.134
The authenticity of host '98.84.168.134 (98.84.168.134)' can't be established.
ED25519 key fingerprint is SHA256:F1I0TeVIAwOvg7TvRQqxKS2d1HtFD0C7JmNlmKC+x4k.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '98.84.168.134' (ED25519) to the list of known hosts.
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@         WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'aws-ec2-key.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "aws-ec2-key.pem": bad permissions
ubuntu@98.84.168.134: Permission denied (publickey).
bubu@fedora-bubu:~/Downloads$ chmod 700 aws-ec2-key.pem
bubu@fedora-bubu:~/Downloads$ ssh -i aws-ec2-key.pem ubuntu@98.84.168.134
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1011-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro
```

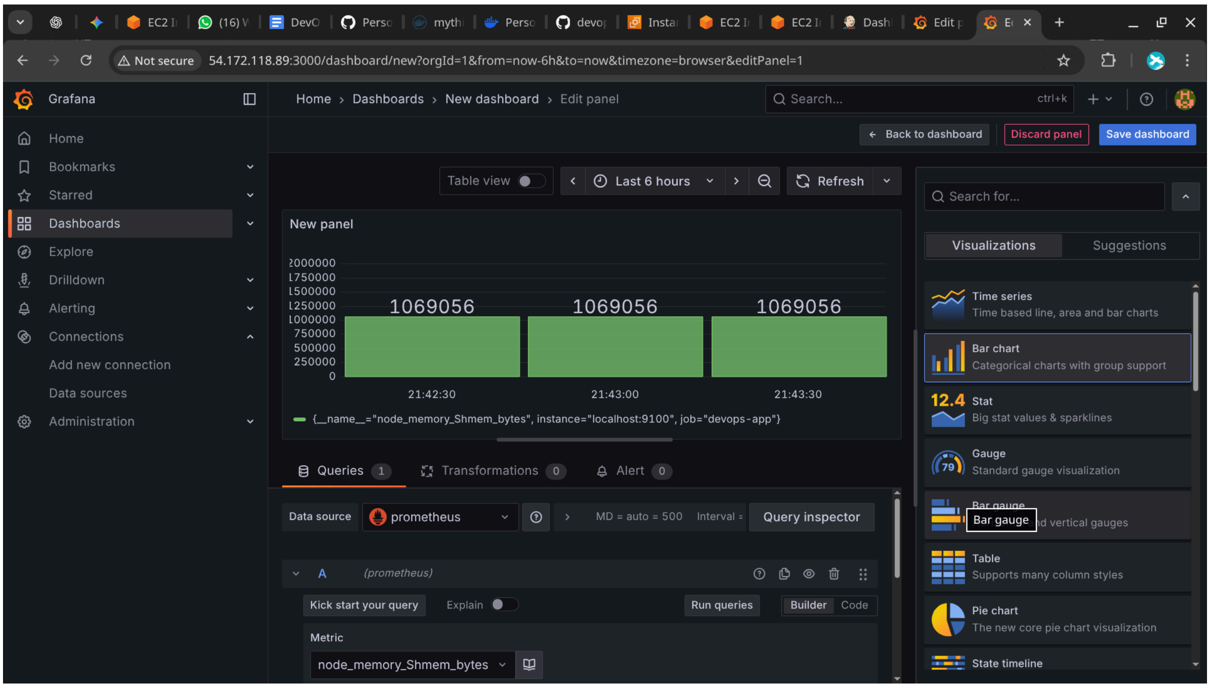



```
root@ip-172-31-24-39:/home/ubuntu# apt install docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base pigz runc ubuntu-fan
Suggested packages:
  ifupdown aufs-tools cgroupfs-mount | cgroup-lite debootstrap docker-buildx docker-compose-v2 docker-doc rinse zfs-fuse | zfsutils
The following NEW packages will be installed:
  bridge-utils containerd dns-root-data dnsmasq-base docker.io pigz runc ubuntu-fan
0 upgraded, 8 newly installed, 0 to remove and 37 not upgraded.
Need to get 79.2 MB of archives.
After this operation, 300 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 pigz amd64 2.8-1 [65.6 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 bridge-utils amd64 1.7.1-1ubuntu2 [33.9 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 runc amd64 1.2.5-0ubuntu1~24.04.1 [8043 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 containerd amd64 1.7.27-0ubuntu1~24.04.1 [37.7 MB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 dns-root-data all 2024071801~ubuntu0.24.04.1 [5918 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 dnsmasq-base amd64 2.90-2ubuntu0.1 [376 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 docker.io amd64 27.5.1-0ubuntu3~24.04.2 [33.0 MB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 ubuntu-fan all 0.12.16+24.04.1 [34.2 kB]
Fetched 79.2 MB in 1s (85.4 MB/s)
Preconfiguring packages ...
Selecting previously unselected package pigz.
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../0-pigz_2.8-1_amd64.deb ...
Unpacking pigz (2.8-1) ...
Selecting previously unselected package bridge-utils.
Preparing to unpack .../1-bridge-utils_1.7.1-1ubuntu2_amd64.deb ...
Unpacking bridge-utils (1.7.1-1ubuntu2) ...
Selecting previously unselected package runc.
Preparing to unpack .../2-runc_1.2.5-0ubuntu1~24.04.1_amd64.deb ...
Unpacking runc (1.2.5-0ubuntu1~24.04.1) ...
Selecting previously unselected package containerd.
Preparing to unpack .../3-containerd_1.7.27-0ubuntu1~24.04.1_amd64.deb ...
```

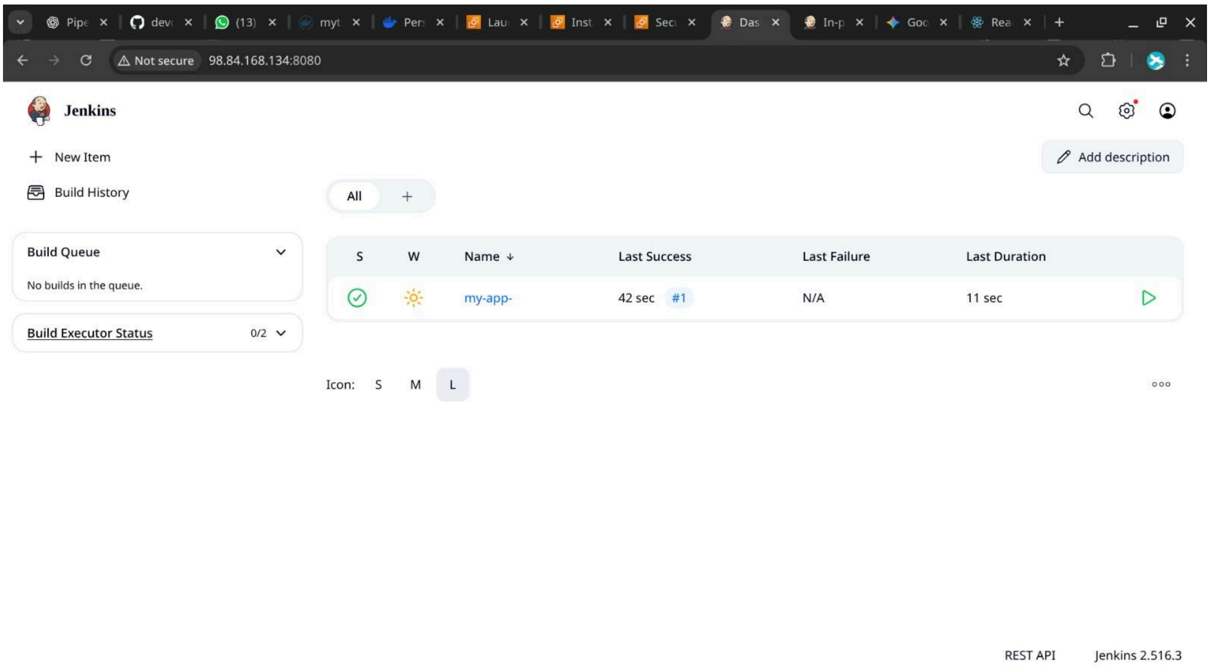
```
build  build.sh  deploy.sh  docker-compose.yml  Dockerfile  Jenkinsfile
bubu@fedora-bubu:~/Desktop/devops-build$ git pull origin dev
From https://github.com/mythili1-14/devops-build
 * branch            dev        -> FETCH_HEAD
Already up to date.
bubu@fedora-bubu:~/Desktop/devops-build$ git add .
bubu@fedora-bubu:~/Desktop/devops-build$ git commit -m "added Jenkinsfile" Jenkinsfile
[dev 3768664] added Jenkinsfile
 1 file changed, 30 insertions(+)
 create mode 100644 Jenkinsfile
bubu@fedora-bubu:~/Desktop/devops-build$ git push origin dev
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 730 bytes | 730.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/mythili1-14/devops-build.git
   45f2575..3768664  dev -> dev
bubu@fedora-bubu:~/Desktop/devops-build$
```

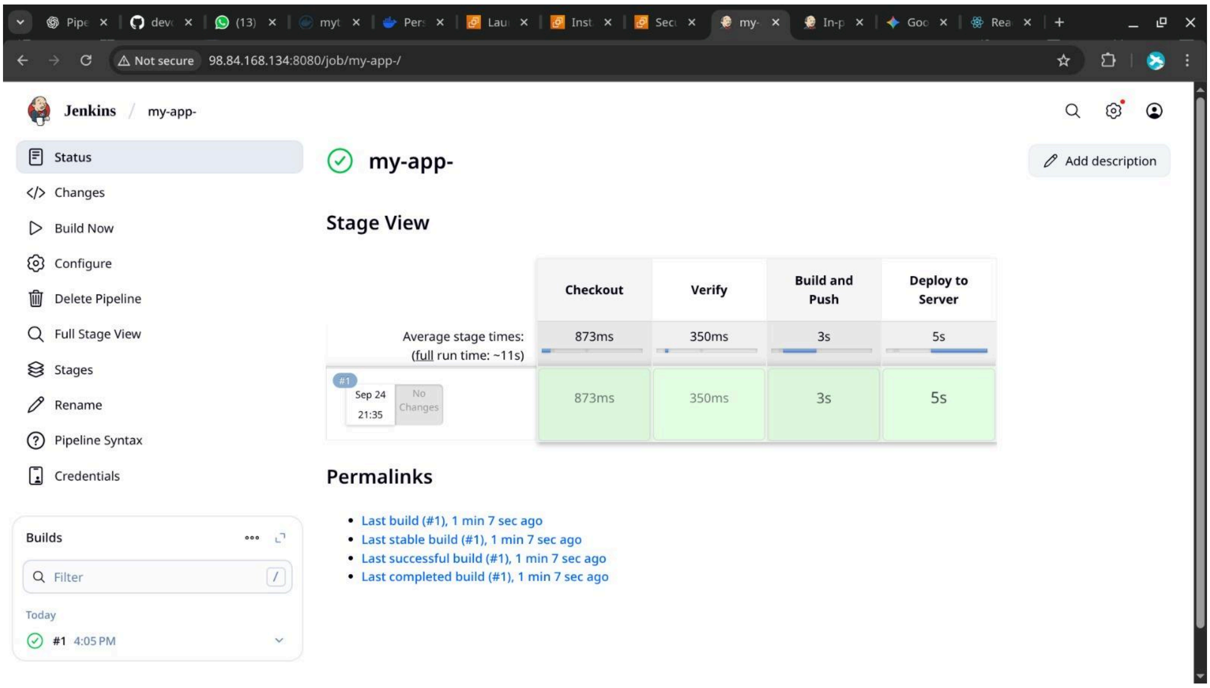
## Step 9: Monitoring

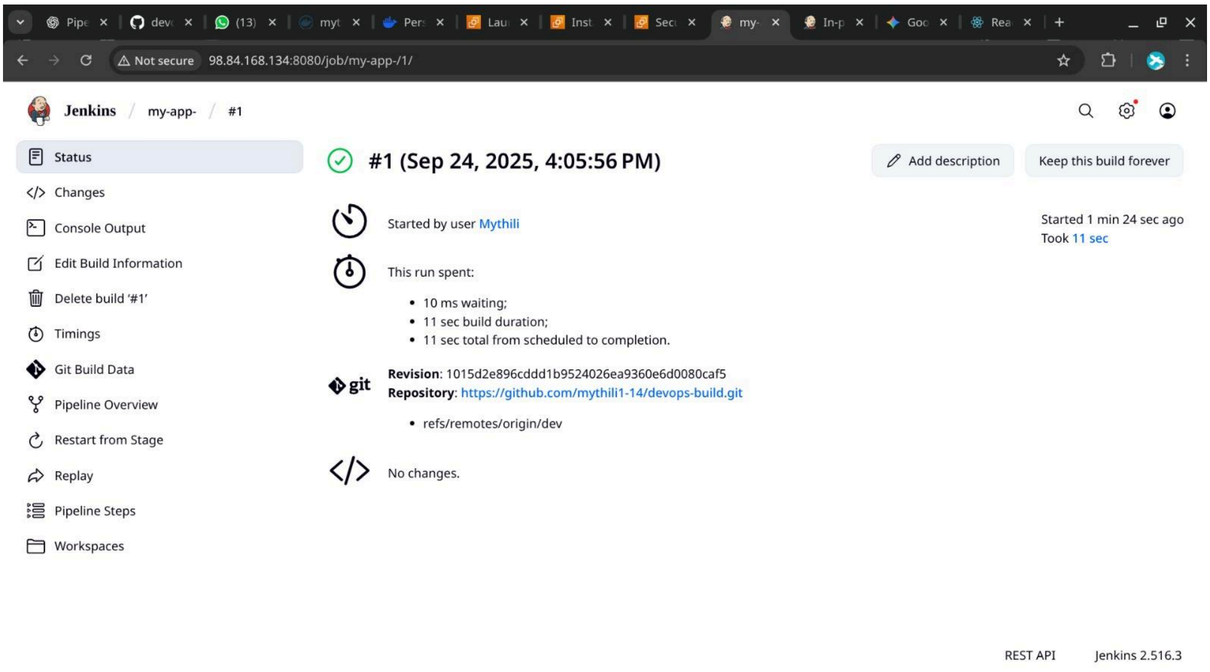
- Install Prometheus to scrape Node Exporter or app metrics, and Grafana for dashboards.
- Add alertmanager for email/Slack alerts when  app is unhealthy



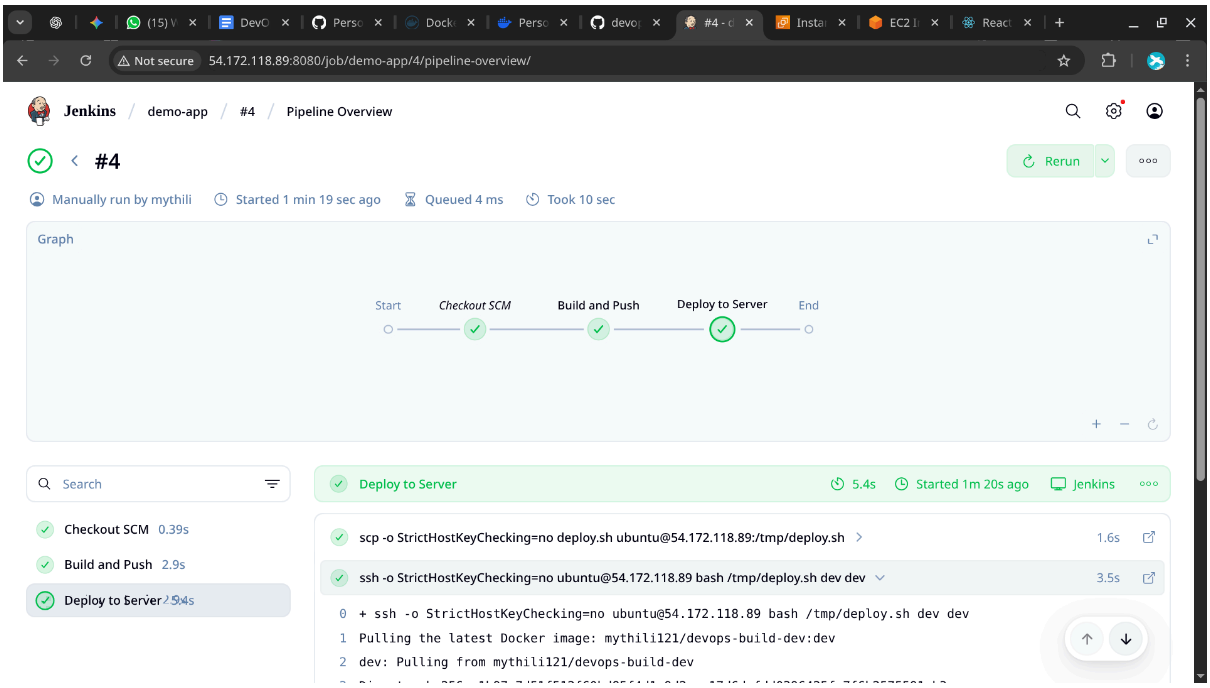
# Step 10 : Output



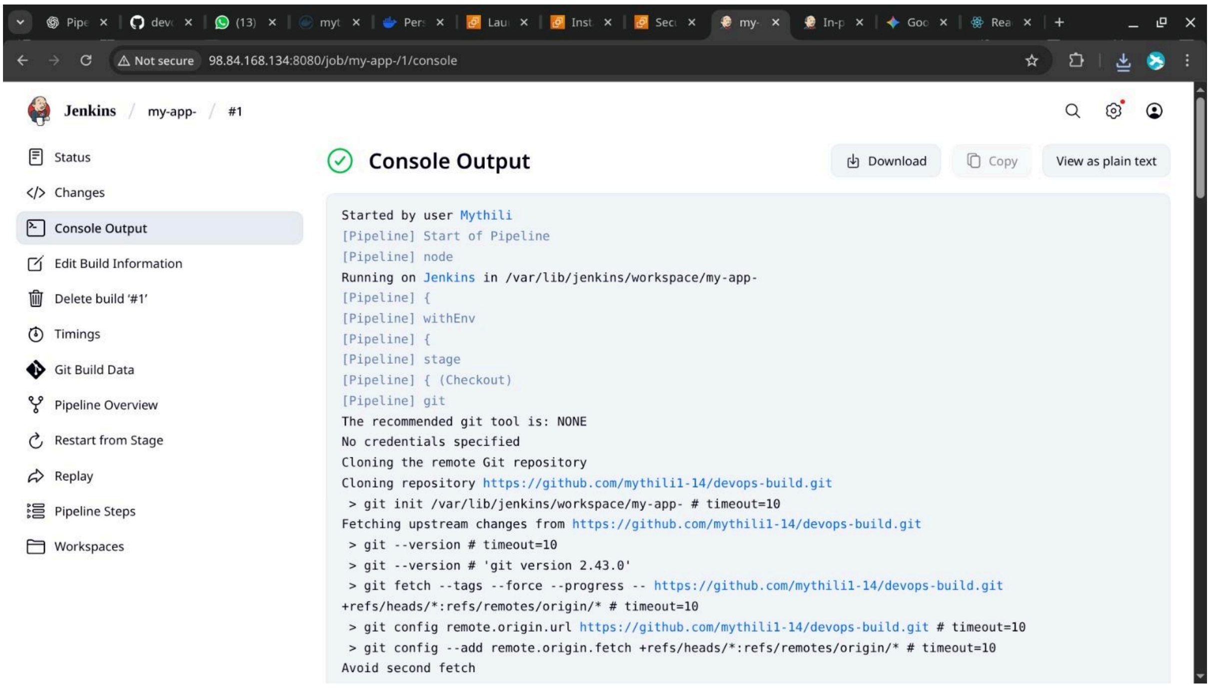



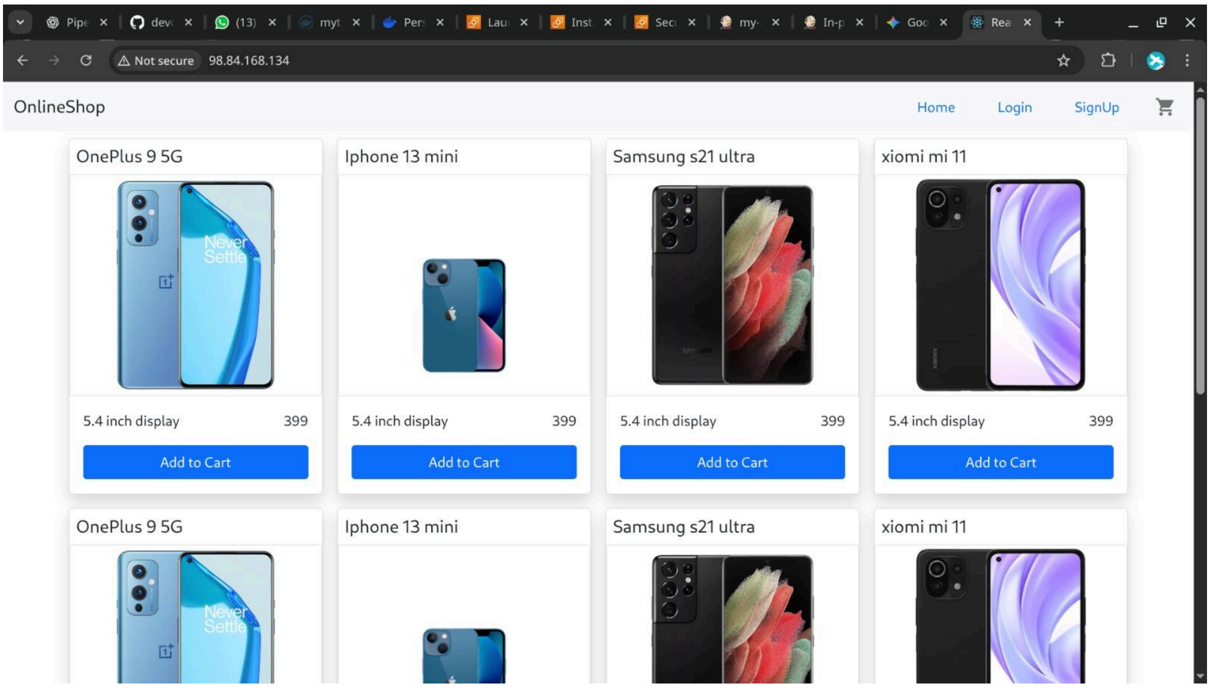


Not secure 98.84.168.134:8080/job/my-app-/1/
Jenkins / my-app- / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
#1 (Sep 24, 2025, 4:05:56 PM)
Add description
Keep this build forever
Started by user Mythili
Started 1 min 24 sec ago
Took 11 sec
This run spent:
10 ms waiting;
11 sec build duration;
11 sec total from scheduled to completion.
git
Revision: 1015d2e896cddd1b9524026ea9360e6d0080caf5
Repository: https://github.com/mythili1-14/devops-build.git
refs/remotes/origin/dev
No changes.
REST API
Jenkins 2.516.3

Not secure 54.172.118.89:8080/job/demo-app/4/pipeline-overview/
Jenkins / demo-app / #4 / Pipeline Overview
#4
Rerun
Manually run by mythili
Started 1 min 19 sec ago
Queued 4 ms
Took 10 sec
Graph
Start
Checkout SCM
Build and Push
Deploy to Server
End
Search
Checkout SCM 0.39s
Build and Push 2.9s
Deploy to Server 5.4s
Deploy to Server
5.4s
Started 1m 20s ago
Jenkins
scp -o StrictHostKeyChecking=no deploy.sh ubuntu@54.172.118.89:/tmp/deploy.sh
1.6s
ssh -o StrictHostKeyChecking=no ubuntu@54.172.118.89 bash /tmp/deploy.sh dev dev
3.5s
0 + ssh -o StrictHostKeyChecking=no ubuntu@54.172.118.89 bash /tmp/deploy.sh dev dev
1 Pulling the latest Docker image: mythili121/devops-build-dev:dev
2 dev: Pulling from mythili121/devops-build-dev


98.84.168.134:8080/job/my-app-/1/console
Jenkins / my-app- / #1
Status
Changes
Console Output
Edit Build Information
Delete build '#1'
Timings
Git Build Data
Pipeline Overview
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console Output
Download
Copy
View as plain text
Started by user Mythili
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/my-app-
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Checkout)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/mythili1-14/devops-build.git
> git init /var/lib/jenkins/workspace/my-app- # timeout=10
Fetching upstream changes from https://github.com/mythili1-14/devops-build.git
> git --version # timeout=10
> git --version # 'git version 2.43.0'
> git fetch --tags --force --progress -- https://github.com/mythili1-14/devops-build.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/mythili1-14/devops-build.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch


98.84.168.134
OnlineShop
Home
Login
SignUp
OnePlus 9 5G
Iphone 13 mini
Samsung s21 ultra
xiomi mi 11
5.4 inch display
399
Add to Cart
5.4 inch display
399
Add to Cart
5.4 inch display
399
Add to Cart
5.4 inch display
399
Add to Cart
OnePlus 9 5G
Iphone 13 mini
Samsung s21 ultra
xiomi mi 11